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30 JUNE 1960
SUPERSEDING
MIL-P-20669
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MILITARY SPECIFICATION

PROTRACTOR SET, SOUND RANGING, PLASTIC, 1 TO 25,000 TIME SCALE INTERVAL

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers a set of two fan-shaped protractors which are graduated in mils, degrees, and a time interval scale on an arc and in yards on one or more of their radii.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-V-351 — Vinyl Chloride Polymer and Copolymer Rigid Sheets.

PPP-B-636 — Box, Fiberboard.

PPP-T-60 — Tape; Pressure-Sensitive Adhesive, Waterproof — for Packaging and Sealing.

PPP-T-97 — Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY

MIL-P-116 — Preservation, Methods of.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

MIL-STD-130 — Identification Marking of U.S. Military Property.

DRAWINGS

CORPS OF ENGINEERS

D 4890-1 — Protractor, Fan, Plastic, Sound Ranging, 4 Sound Seconds.

FSC 6675

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D 4891–1 — Protractor, Fan, Plastic, Sound Ranging, 2 Sound Seconds.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addresed to the Official Classification Committee, 1 Park Ave. at 33 St., New York 16, N. Y.)

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Association, 1424 16th St., N. W., Washington, D. C.)

3. REQUIREMENTS

3.1 Description. Each protractor set shall consist of a 4-sound-second protractor and a 2-sound-second protractor as shown on Drawings D 4890–1 and D 4891–1, respectively, and as specified herein.

3.1.1 The drawings forming a part of this specification are engineering design drawings and are not intended for use as shop drawings. Critical dimensions and tolerances are furnished; however, the contractor is responsible for preparing his own shop drawings. Where tolerances are not shown or where those prescribed could cumulatively result in improper fits, the contractor is responsible for providing tolerances in accordance with industry standards for similar equipment and within those prescribed on the engineering design drawings to insure fit, assembly, and operation of the equipment. No deviation from dimensions or tolerances shown on the engineering design drawings is permissible without prior written approval of the contracting officer.

3.2 Preproduction model. The preproduction model is defined as a production model which will be identical to the product which the manufacturer will subsequently produce in fulfillment of the contract. As soon as practicable after award of the contract or order and prior to submission of any complete units for final acceptance, the contractor shall furnish a preproduction model for examination and test to determine conformance to this specification. Examination and tests shall be those specified herein. Approval of the preproduction model by the activity concerned shall not relieve the contractor of his obligation to supply equipment conforming to this specification. Any changes or deviations of production units from the preproduction model shall be subject to the approval of the contracting officer.

3.3 Material. Material shall be as specified herein and as shown on the applicable drawings.

3.3.1 *Plastic.* The plastic material shall conform to Specification L–V–351, class 1. The material shall be free from residual solvents.

3.4 Temperature conditions. The protractors shall not be damaged by storage at any ambient temperature from 155°F. for a period of not less than 4 hours to minus 65°F. for a period of not less than 3 days. There shall be no visible deterioration or clouding of the protractors and the protractors shall remain flat with no point more than 0.0075 inch from a true flat plane.

3.5 Zero range point. The notched end of each protractor shall have a hole located at the zero range point and at the center of curvature of the graduated arcs within a

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tolerance of 0.005 inch in any direction. The notch shall be milled to the hole so that the sides of the notch shall be tangent to the hole, leaving no shoulder.

3.6 Graduations and markings. Graduations, numerals, and other markings shall be printed on the underside of the protractor so as to be readable from the upper side. Lettering and numbering shall be of the size and style shown on the drawings.

3.6.1 *Scales.* Range scales and subbase scales shall be divided into equal graduations with the origin of each scale at the zero range point. Degree and mil scales shall be true (1,600 mils equal 90°) with the origin of each scale at the edge of the fan. The time interval scales shall be divided according to the equation:

$$\frac{t}{s} = \text{Sin } \theta$$

where:

t = time interval (seconds) corresponding to the graduation,

s = number of sound seconds indicated in the description title of the fan, and

θ = the angle as measured at the center of the hole at the notched end of the fan from the origin of the time interval to the graduation.

Tabulation of the values of the angle θ, determined from the above equation, is given in table I. The origin of each time interval scale shall be at the edge of the fan. The width of graduation lines, tolerance in their width, and tolerance in their position shall be as shown on the drawings.

3.6.2 *Markings.* Figures and other markings shall be printed in black, except as noted on the drawings.

3.7 Protective coating. To protect the printing from abrasion, the printing shall be covered with a transparent film of lacquer or a sheet of the same material as the main part of the protractor. Lacquer, if used, shall not impair the transparency of the protractor. Cover sheets, if used, shall be attached by a laminating or compositing process in such a manner that the plastic sheets are permanently united. The compositing process shall not impair the transparency of the finished protractor. The protractors shall show no sign of separation of the protective film or cover.

3.8 Registration holes. At the option of the contractor and if required in the manufacturing process, registration holes may be drilled or punched through the protractors. Registration holes shall not exceed four in number and shall have a maximum diameter of ⅜ inch. The registration holes shall be at least ½ inch from any edge of the protractor and shall be located so as not to interfere with any scales, markings, template holes, or with the notch at the lower end of the protractor.

3.9 Identification marking. Each protractor shall be identified in accordance with Standard MIL–STD–130.

3.10 Finishing. All surfaces and edges of the protractor shall be true and shall have a smoothness equivalent to that obtained by milling and subsequent polishing. The polishing operation shall not impair the accuracy of the protractor.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to per-

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form any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection of the protractors.

(b) Acceptance inspection of the protractors.

(c) Acceptance inspection of preparation for delivery.

4.3 Preproduction inspection.

4.3.1 *Examination.* The protractor shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

Major

101. Material not as specified.

102. Zero range point not at center of curvature of the graduated circular arcs.

103. Graduation tolerances not within specified limits.

104. Markins, graduations, scales, and numbers not as specified.

105. Registration holes, if present, not as specified.

106. Dimensions not as specified.

Minor

201. Improper identification.

202. Finishing not as specified.

4.3.2 *Tests.* The protractor shall be tested as follows. Failure to pass any test shall be cause for rejection.

4.3.2.1 *Temperature conditions.* Protractors shall be placed on a flat surface and subjected to a cycle of temperatures in the following order:

(a) A high-temperature series, consisting of three exposures of 4 hours each on 3 successive days at a temperature of 155°F. and a relative humidity of 15 percent.

(b) A low-temperature condition, consisting of 72 consecutive hours of exposure at a temperature of minus 65°F.

Upon completion of this cycle, the protractors shall be examined. Evidence of deterioration or clouding shall constitute failure. Warping shall be checked by placing the protractors on a flat surface and by measuring the deviation by use of a feeler gage. Evidence of flatness more than 0.0075 inch from a true flat plane shall constitute failure.

4.3.2.2 *Protective coating.* A set of protractors shall be immersed for 6 hours in water at a temperature of 75° plus or minus 2°F. Immediately thereafter, the protractors shall be bent in the lengthwise and crosswise directions through a circular arc of 45° over a mandrel 6 inches in diameter. While the protractors are being bent and after they have been straightened, they shall be examined. Evidence of signs of separation of the protective film or cover from the protractor shall constitute failure.

4.4 Acceptance inspection.

4.4.1 *Sampling.*

4.4.1.1 *For examination.* Sampling shall be conducted in accordance with Standard MIL–STD–105.

4.4.1.2 *For tests.* Sampling shall be conducted in accordance with Standard MIL–STD–105, inspection level L–4.

4.4.2 *Examination.*

4.4.2.1 *Sample examination.* Samples selected in accordance with 4.4.1.1 shall be examined for defects listed in 4.3.1. AQL for

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major defects shall be 2.5 percent defective and for minor defects 4.0 percent defective.

4.4.3 *Tests.*

4.4.3.1 *Sample tests.* Samples selected in accordance with 4.4.1.2 shall be tested as specified in 4.3.2. AQL shall be 10 percent defective.

4.5 Acceptance inspection of preparation for delivery. The preservation, packaging, packing, and marking shall be examined and tested for compliance with section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.* Each set of protractors shall be preserved in accordance with Specification MIL–P–116, method IC–1.

5.1.2 *Level C.* The protractors shall be preserved and packaged to afford protection against deterioration and damage.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* The protractors shall be packed in close-fitting fiber boxes conforming to Specification PPP–B–636, type I, class 2, compliance symbol V3c, style optional. After closure of the boxes, all joints and seams of the boxes shall be covered with minimum 3-inch-wide tape conforming to Specification PPP–T–60, type II, class 1. The boxes shall be reinforced with bands of tape conforming to Specification PPP–T–97, type IV, width as applicable, applied in accordance with the appendix to the tape specification. Strapping shall not be required.

5.2.2 *Level B.* The protractors shall be packed in boxes conforming to Specification PPP–B–636, type I, class 1, compliance symbol and style optional.

5.2.3 *Level C.* The protractors shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with Uniform Freight Classification Rules or with National Motor Freight Classification Rules or with other carrier rules applicable to the mode of transportation.

5.3 Marking. Packages and shipping containers shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 Intended use. The protractors are used to locate the origin of sounds such as gun reports.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Administrative provisions for inspection records (see 4.1).
(c) Levels of preservation and packaging and level of packing required (see 5.1 and 5.2).
(d) There are no level B preservation and packaging requirements.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to tbe regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacturer, use, or sell any patented invention that may in any be related thereto.

Custodians:
Army—Corps of Engineers
Navy—Marine Corps

Preparing activity:
Army—Corps of Engineers

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TABLE I. Θ *Values of* Θ *and tan* Θ *for 2- and 4-sound-second protractor*

t(2)	t(4)	Θ	tan Θ	t(2)	t(4)	Θ	tan Θ
	0	0	0.000000	0.27	0.54	7°45′31″	0.136247
	0.01	0°08′36″	0.002500		0.55	7°54′12″	0.138819
0.01	0.02	0°17′11″	0.005000	0.28	0.56	8°02′52″	0.141393
	0.03	0°25′47″	0.007500		0.57	8°11′33″	0.143969
0.02	0.04	0°34′23″	0.010001	0.29	0.58	8°20′14″	0.146549
	0.05	0°42′58″	0.012501		0.59	8°28′55″	0.149131
0.03	0.06	0°51′34″	0.015002	0.30	0.60	8°37′37″	0.151717
	0.07	1°00′10″	0.017503		0.61	8°46′19″	0.154305
0.04	0.08	1°08′46″	0.020004	0.31	0.62	8°55′00″	0.156896
	0.09	1°17′21″	0.022506		0.63	9°03′43″	0.159491
0.05	0.10	1°25′57″	0.025008	0.32	0.64	9°12′25″	0.162088
	0.11	1°34′33″	0.027510		0.65	9°21′07″	0.164689
0.06	0.12	1°43′09″	0.030014	0.33	0.66	9°29′50″	0.167293
	0.13	1°51′45″	0.032517		0.67	9°38′33″	0.169900
0.07	0.14	2°00′21″	0.035021	0.34	0.68	9°47′16″	0.172511
	0.15	2°08′57″	0.037526		0.69	9°56′00″	0.175125
0.08	0.16	2°17′33″	0.040032	0.35	0.70	10°04′43″	0.177743
	0.17	2°26′09″	0.042538		0.71	10°13′27″	0.180364
0.09	0.18	2°34′45″	0.045046	0.36	0.72	10°22′11″	0.182989
	0.19	2°43′21″	0.047554		0.73	10°30′55″	0.185617
0.10	0.20	2°51′58″	0.050063	0.37	0.74	10°39′40″	0.188249
	0.21	3°00′34″	0.052573		0.75	10°48′25″	0.190885
0.11	0.22	3°09′10″	0.055083	0.38	0.76	10°57′10″	0.193525
	0.23	3°17′47″	0.057595		0.77	11°05′55″	0.196169
0.12	0.24	3°26′23″	0.060108	0.39	0.78	11°14′41″	0.198817
	0.25	3°35′00″	0.062622		0.79	11°23′27″	0.201468
0.13	0.26	3°43′37″	0.065138	0.40	0.80	11°32′13″	0.204124
	0.27	3°52′13″	0.067654		0.81	11°40′59″	0.206784
0.14	0.28	4°00′50″	0.070172	0.41	0.82	11°49′46″	0.209448
	0.29	4°09′27″	0.072691		0.83	11°58′33″	0.212117
0.15	0.30	4°18′04″	0.075212	0.42	0.84	12°07′20″	0.214790
	0.31	4°26′42″	0.077734		0.85	12°16′08″	0.217467
0.16	0.32	4°35′19″	0.080257	0.43	0.86	12°24′56″	0.220148
	0.33	4°43′56″	0.082782		0.87	12°33′44″	0.222835
0.17	0.34	4°52′34″	0.085309	0.44	0.88	12°42′33″	0.225525
	0.35	5°01′11″	0.087837		0.89	12°51′21″	0.228221
0.18	0.36	5°09′49″	0.090367	0.45	0.90	13°00′10″	0.230921
	0.37	5°18′27″	0.092898		0.91	13°09′00″	0.233626
0.19	0.38	5°27′05″	0.095432	0.46	0.92	13°17′49″	0.236336
	0.39	5°35′43″	0.097967		0.93	13°26′39″	0.239051
0.20	0.40	5°44′21″	0.100504	0.47	0.94	13°35′30″	0.241771
	0.41	5°52′59″	0.103043		0.95	13°44′21″	0.244496
0.21	0.42	6°01′38″	0.105584	0.48	0.96	13°53′12″	0.247226
	0.43	6°10′16″	0.108127		0.97	14°02′03″	0.249961
0.22	0.44	6°18′55″	0.110672	0.49	0.98	14°10′55″	0.252702
	0.45	6°27′34″	0.113219		0.99	14°19′47″	0.255448
0.23	0.46	6°36′13″	0.115768	0.50	1.00	14°28′39″	0.258199
	0.47	6°44′52″	0.118320		1.01	14°37′32″	0.260956
0.24	0.48	6°53′32″	0.120873	0.51	1.02	14°46′25″	0.263718
	0.49	7°02′11″	0.123430		1.03	14°55′18″	0.266486
0.25	0.50	7°10′51″	0.125988	0.52	1.04	15°04′12″	0.269260
	0.51	7°19′31″	0.128549		1.05	15°13′06″	0.272040
0.26	0.52	7°28′11″	0.131113	0.53	1.06	15°22′01″	0.274825
	0.53	7°36′51″	0.133679		1.07	15°30′56″	0.277617

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t(2)	t(4)	θ	tan θ	t(2)	t(4)	θ	tan θ
0.54	1.08	15°39′51″	0.280415	0.82	1.64	24°12′17″	0.449519
	1.09	15°48′47″	0.283218		1.65	24°21′43″	0.452820
0.55	1.10	15°57′43″	0.286028	0.83	1.66	24°31′10″	0.456133
	1.11	16°06′40″	0.288844		1.67	24°40′37″	0.459459
0.56	1.12	16°15′37″	0.291667	0.84	1.68	24°50′05″	0.462798
	1.13	16°24′34″	0.294496		1.69	24°59′33″	0.466149
0.57	1.14	16°33′32″	0.297331	0.85	1.70	25°09′02″	0.469513
	1.15	16°42′30″	0.300173		1.71	25°18′32″	0.472890
0.58	1.16	16°51′29″	0.303022	0.86	1.72	25°28′03″	0.476281
	1.17	17°00′28″	0.305877		1.73	25°37′35″	0.479685
0.59	1.18	17°09′27″	0.308740	0.87	1.74	25°47′07″	0.483102
	1.19	17°18′27″	0.311609		1.75	25°56′40″	0.486534
0.60	1.20	17°27′27″	0.314485	0.88	1.76	26°06′14″	0.489979
	1.21	17°36′28″	0.317369		1.77	26°15′49″	0.493438
0.61	1.22	17°45′29″	0.320260	0.89	1.78	26°25′24″	0.496912
	1.23	17°54′31″	0.323158		1.79	26°35′00″	0.500400
0.62	1.24	18°03′33″	0.326063	0.90	1.80	26°44′37″	0.503903
	1.25	18°12′36″	0.328976		1.81	26°54′15″	0.507421
0.63	1.26	18°21′39″	0.331896	0.91	1.82	27°03′54″	0.510954
	1.27	18°30′42″	0.334824		1.83	27°13′33″	0.514502
0.64	1.28	18°39′47″	0.337760	0.92	1.84	27°23′14″	0.518065
	1.29	18°48′51″	0.340704		1.85	27°32′55″	0.521644
0.65	1.30	18°57′56″	0.343656	0.93	1.86	27°42′37″	0.525239
	1.31	19°07′02″	0.346615		1.87	27°52′20″	0.258850
0.66	1.32	19°16′08″	0.349583	0.94	1.88	28°02′03″	0.532477
	1.33	19°25′14″	0.352559		1.89	28°11′48″	0.536121
0.67	1.34	19°34′21″	0.355544	0.95	1.90	28°21′34″	0.539781
	1.35	19°43′29″	0.358537		1.91	28°31′20″	0.543459
0.68	1.36	19°52′37″	0.361538	0.96	1.92	28°41′07″	0.547153
	1.37	20°01′45″	0.364549		1.93	28°50′56″	0.550865
0.69	1.38	20°10′54″	0.367568	0.97	1.94	29°00′45″	0.554594
	1.39	20°20′04″	0.370596		1.95	29°10′35″	0 558341
0.70	1.40	20°29′14″	0.373632	0.98	1.96	29°20′26″	0.562106
	1.41	20°38′25″	0.376678		1.97	29°30′18″	0.565889
0.71	1.42	20°47′36″	0.379733	0.99	1.98	29°40′11″	0.569690
	1.43	20°56′48″	0.382798		1.99	29°50′05″	0.573511
0.72	1.44	21°06′01″	0.385872	1.00	2.00	30°00′00″	0.577350
	1.45	21°15′14″	0.388955		2.01	30°09′56″	0.581209
0.73	1.46	21°24′27″	0.392048	1.01	2.02	30°19′53″	0.585087
	1.47	21°33′41″	0.395151		2.03	30°29′51″	0 588985
0.74	1.48	21°42′56″	0.398264	1.02	2.04	30°39′50″	0.592903
	1.49	21°52′12″	0.401387		2.05	30°49′50″	0.596841
0.75	1.50	22°01′28″	0.404520	1.03	2.06	30°59′51″	0.600800
	1.51	22°10′44″	0.407663		2.07	31°09′53″	0.604780
0.76	1.52	22°20′01″	0.410817	1.04	2.08	31°19′56″	0.608781
	1.53	22°29′19″	0.413981		2.09	31°30′00″	0.612803
0.77	1.54	22°38′37″	0.417156	1.05	2.10	31°40′06″	0.616847
	1.55	22°47′57″	0.420341		2.11	31°50′12″	0 620913
0.78	1.56	22°57′16″	0.423538	1.06	2.12	32°00′20″	0.625002
	1.57	23°06′37″	0.426745		2.13	32°10′28″	0.629113
0.79	1.58	23°15′58″	0.429964	1.07	2.14	32°20′38″	0.633247
	1.59	23°25′19″	0.433194		2.15	32°30′49″	0 637404
0.80	1.60	23°34′41″	0.436436	1.08	2.16	32°41′01″	0.641585
	1.61	23°44′04″	0.439689		2.17	32°51′14″	0.645790
0.81	1.62	23°53′28″	0.442954	1.09	2.18	33°01′29″	0.650020
	1.63	24°02′52″	0.446230		2.19	33°11′44″	0.654274

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t(2)	t(4)	θ	tan θ	t(2)	t(4)	θ	tan θ
1.10	2.20	33°22′01″	0.658553	1.38	2.76	43°37′48″	0.953290
	2.21	33°32′19″	0.662857		2.77	43°49′42″	0.959915
1.11	2.22	33°42′39″	0.667187	1.39	2.78	44°01′38″	0.966607
	2.23	33°52′59″	0.671544		2.79	44°13′36″	0.973367
1.12	2.24	34°03′21″	0.675926	1.40	2.80	44°25′37″	0.980196
	2.25	34°13′44″	0.680336		2.81	44°37′41″	0.987096
1.13	2.26	34°24′08″	0.684773	1.41	2.82	44°49′46″	0.994068
	2.27	34°34′34″	0.689238		2.83	45°01′55″	1.001113
1.14	2.28	34°45′01″	0.693731	1.42	2.84	45°14′06″	1.008234
	2.29	34°55′29″	0.698252		2.85	45°26′19″	1.015432
1.15	2.30	35°05′59″	0.702802	1.43	2.86	45°38′36″	1.022708
	2.31	35°16′30″	0.707382		2.87	45°50′54″	1.030064
1.16	2.32	35°27′02″	0.711991	1.44	2.88	46°03′16″	1.037503
	2.33	35°37′36″	0.716631		2.89	46°15′41″	1.045025
1.17	2.34	35°48′11″	0.721302	1.45	2.90	46°28′08″	1.052633
	2.35	35°58′47″	0.726004		2.91	46°40′38″	1.060329
1.18	2.36	36°09′25″	0.730738	1.46	2.92	46°53′11″	1.068115
	2.37	36°20′05″	0.735504		2.93	47°05′47″	1.075992
1.19	2.38	36°30′45″	0.740302	1.47	2.94	47°18′26″	1.083964
	2.39	36°41′28″	0.745134		2.95	47°31′08″	1.092031
1.20	2.40	36°52′12″	0.750000	1.48	2.96	47°43′53″	1.100197
	2.41	37°02′57″	0.754900		2.97	47°56′41″	1.108464
1.21	2.42	37°13′44″	0.759835	1.49	2.98	48°09′33″	1.116834
	2.43	37°24′32″	0.764805		2.99	48°22′27″	1.125309
1.22	2.44	37°35′22″	0.769812	1.50	3.00	48°35′25″	1.133893
	2.45	37°46′14″	0.774855		3.01	48°48′27″	1.142589
1.23	2.46	37°57′07″	0.779935	1.51	3.02	49°01′31″	1.151398
	2.47	38°08′02″	0.785053		3.03	49°14′39″	1.160324
1.24	2.48	38°18′58″	0.790210	1.52	3.04	49°27′51″	1.169369
	2.49	38°29′56″	0.795405		3.05	49°41′06″	1.178538
1.25	2.50	38°40′56″	0.800641	1.53	3.06	49°54′25″	1.187832
	2.51	38°51′57″	0.805917		3.07	50°07′48″	1.197256
1.26	2.52	39°03′00″	0.811234	1.54	3.08	50°21′14″	1.206814
	2.53	39°14′05″	0.816592		3.09	50°34′44″	1.216507
1.27	2.54	39°25′12″	0.821994	1.55	3.10	50°48′18″	1.226341
	2.55	39°36′20″	0.827438		3.11	51°01′56″	1.236319
1.28	2.56	39°47′31″	0.832927	1.56	3.12	51°15′38″	1.246445
	2.57	39°58′43″	0.838460		3.13	51°29′24″	1.256724
1.29	2.58	40°09′56″	0.844039	1.57	3.14	51°43′14″	1.267160
	2.59	40°21′12″	0.849665		3.15	51°57′09″	1.277756
1.30	2.60	40°32′30″	0.855337	1.58	3.16	52°11′08″	1.288519
	2.61	40°43′49″	0.861058		3.17	52°25′11″	1.299453
1.31	2.62	40°55′11″	0.866828	1.59	3.18	52°39′19″	1.310563
	2.63	41°06′34″	0.872647		3.19	52°53′31″	1.321855
1.32	2.64	41°18′00″	0.878518	1.60	3.20	53°07′48″	1.333333
	2.65	41°29′27″	0.884440		3.21	53°22′10″	1.345005
1.33	2.66	41°40′56″	0.890414	1.61	3.22	53°36′37″	1.356875
	2.67	41°52′28″	0.896443		3.23	53°51′09″	1.368951
1.34	2.68	42°04′01″	0.902526	1.62	3.24	54°05′45″	1.381239
	2.69	42°15′37″	0.908665		3.25	54°20′27″	1.393746
1.35	2.70	42°27′15″	0.914860	1.63	3.26	54°35′15″	1.406480
	2.71	42°38′55″	0.921114		3.27	54°50′07″	1.419447
1.36	2.72	42°50′37″	0.927426	1.64	3.28	55°05′05″	1.432656
	2.73	43°02′22″	0.933799		3.29	55°20′09″	1.446115
1.37	2.74	43°14′08″	0.940233	1.65	3.30	55°35′19″	1.459834
	2.75	43°25′57″	0.946729		3.31	55°50′34″	1.473821

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t(2)	t(4)	θ	tan θ	t(2)	t(4)	θ	tan θ
1.66	3.32	56°05′55″	1.488086	1.70	3.40	58°12′42″	1.613569
	3.33	56°21′23″	1.502640		3.41	58°29′05″	1.630870
1.67	3.34	56°36′57″	1.517492	1.71	3.42	58°45′35″	1.648580
	3.35	56°52′37″	1.532655		3.43	59°02′13″	1.666719
1.68	3.36	57°08′24″	1.548141	1.72	3.44	59°19′00″	1.685303
	3.37	57°24′18″	1.563961		3.45	59°35′54″	1.704353
1.69	3.38	57°40′19″	1.580129	1.73	3.46	59°52′58″	1.723891
	3.39	57°56′27″	1.596660				

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